Printer & Scanner Basics

Introduction to Printers

- Printers are commonly used output devices that produce a hard copy of document stored in electronic form, i.e they put information from computer on to paper.
- There are various kinds of printers available today like Impact printers, Bubble-jet printers, Laser printers, Thermal printers etc.
- **Types of printers**
  - Impact
    - Daisy-Wheel
    - Dot-Matrix
  - Solid-Ink Printers
  - Thermal Printers
  - Inkjet Printers
  - Laser Printers
  - 3D Printers

Daisy-Wheel Printers

- works on the same principle as ball-head typewriter
- consists of a disk made of plastic or metal on which characters stand out along the outer edge.
- The printer rotates the disk to print a character until the desired letter is facing the paper, after which a hammer called solenoid strikes forcing the character to hit an ink ribbon making a mark of the character on the paper.
- speed is rated by cps (number of characters per second).
- main disadvantage is that they make noise when printing and these kind of printers cannot print graphics.
- advantage is that they are not expensive and can produce letter-quality text.

Fig a: Daisy-wheel

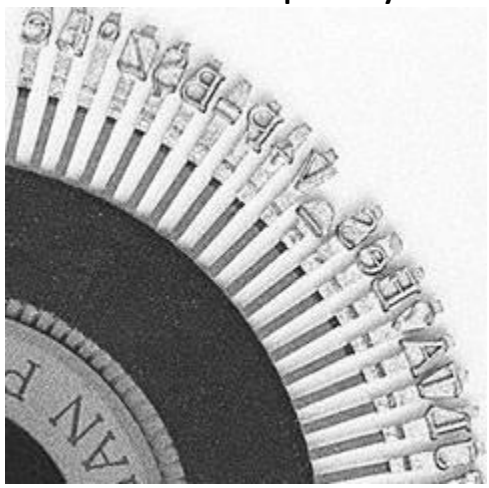


Fig b: A section in detail

Dot-Matrix Printers

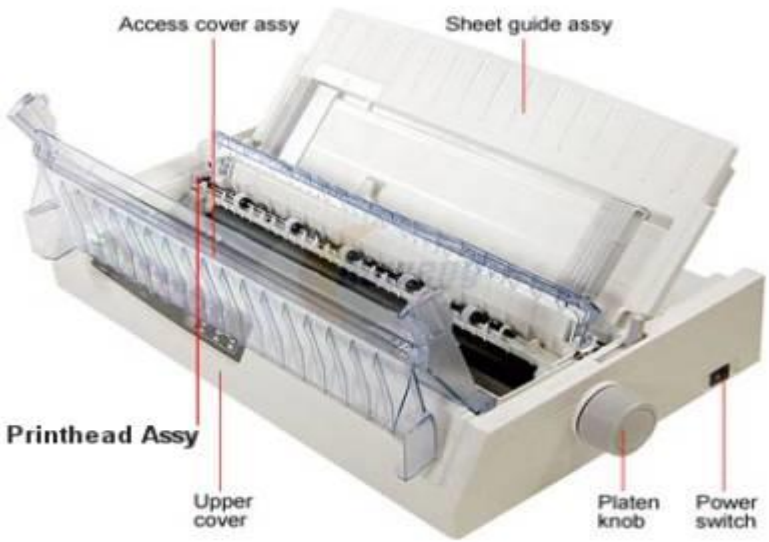


Fig.: Dot-matrix printer

Fig.: Continuous feed paper

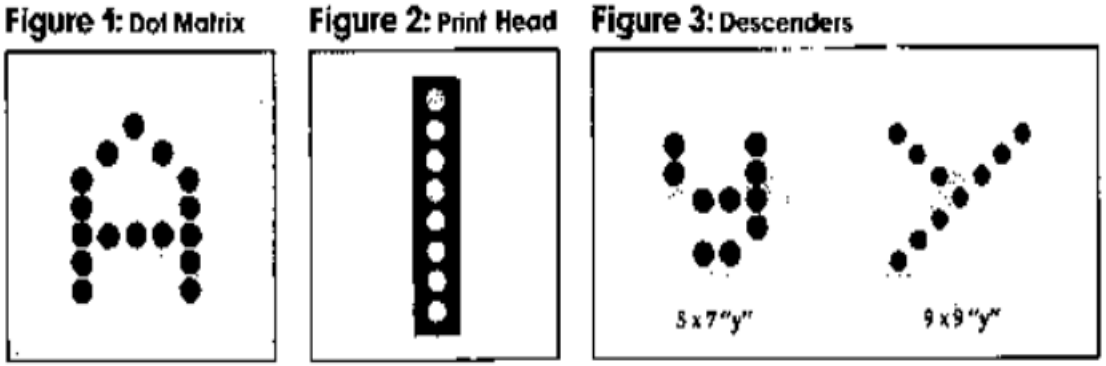


Figure: 8-pin dot-matrix print head with examples of printed letters.

Dot-Matrix Printers

- **Advantages of Dot-matrix printers include the following:**
  - They can print on multi-part stationary or carbon copies
  - Lower printing costs compared with Inkjet or Laser, the reason being that dot-matrix printers use a ribbon and rugged printing process.
  - These can withstand unclean or dusty environment whereas Inkject or laser jet printers require clean environment
  - Suitable for traction fed paper or continuous paper feed. Sometimes you may need to print an activity using continuous paper feed. Inkjet and Laser printers use discrete sheets of paper, and normally do not use continuous paper feed.
  - Using these printers require negligible operator training.
  - These printers usually cost less (initial purchase cost may be more, but running cost is negligible)
- **The main disadvantages of Dot-matrix printers are:**
  - Dot-matrix printers are noisy but not as much as Daisy-wheel printers
  - The quality of print is not as good as laser and inkjet printers.
  - cannot be used for producing good quality images or for photo printing.
  - These printers are slow compared with laser printers.

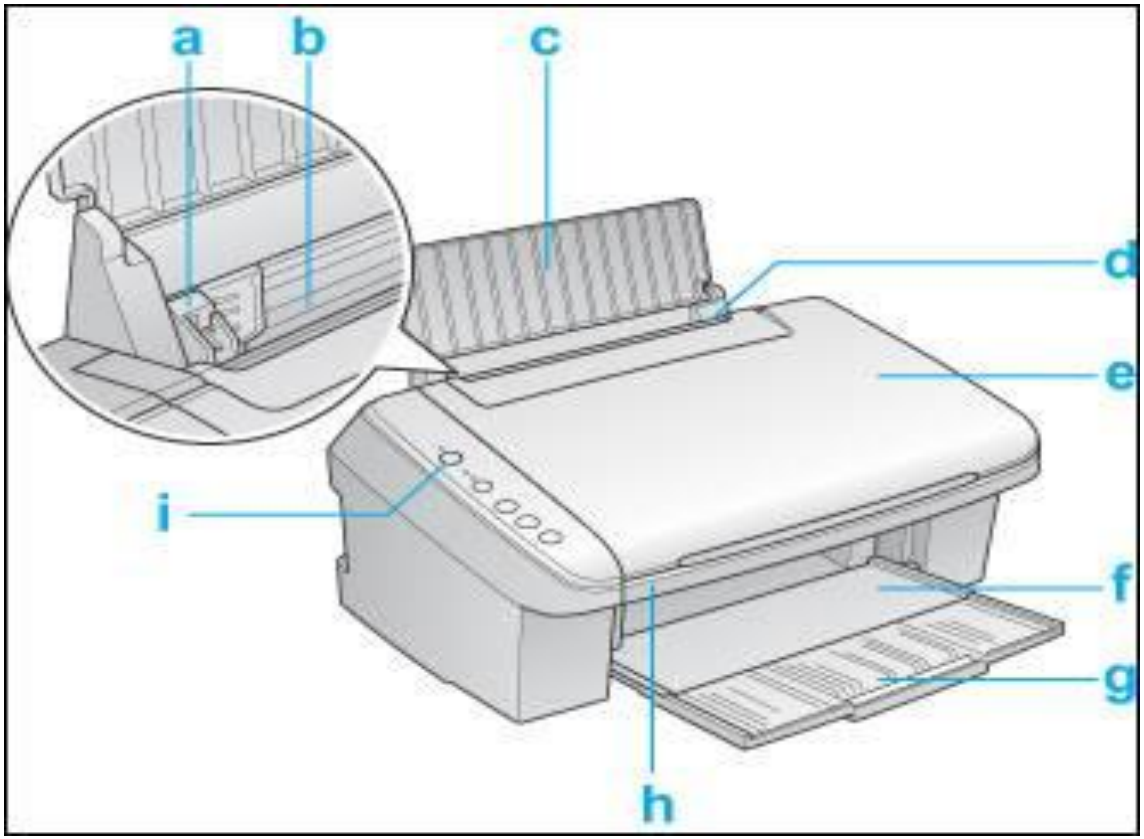
Solid-Ink Printers

- Solid-Ink printers use ink in a waxy solid form than liquid form which avoids problems like spillage.
- And these kinds of printers print one line at a time and these printers are best suitable for graphic companies that need true color at a price lower than a color laser printer.
- **Advantages and Disadvantages**
  - Advantages are good print quality, ease of use and generate less waste
  - Solid ink printers are less sensitive to paper thickness and paper fibers. This allows printing on a number of different surfaces including recycled and handmade paper.
  - disadvantages include more power consumption and odour of wax

Thermal Printers

- **Direct Thermal printer**
  - Direct Thermal printer use a heated printhead to burn dots into the surface of special heat-sensitive paper.
  - It is similar to older fax machines. The disadvantage in this is that the paper that is used gets darken early, thus making difficult to read.
- **Thermal wax-transfer printer**
  - Thermal wax-transfer printers use a heat-sensitive-ribbon instead of heat-sensitive paper.
  - Thermal printhead melts wax-based ink from the ribbon on to the paper.
  - The disadvantages with thermal transfer printers are that the heating and cooling of the print head determines their speed which means that since the printhead is extensively used, it has to be replaced often.

Inkjet Printers



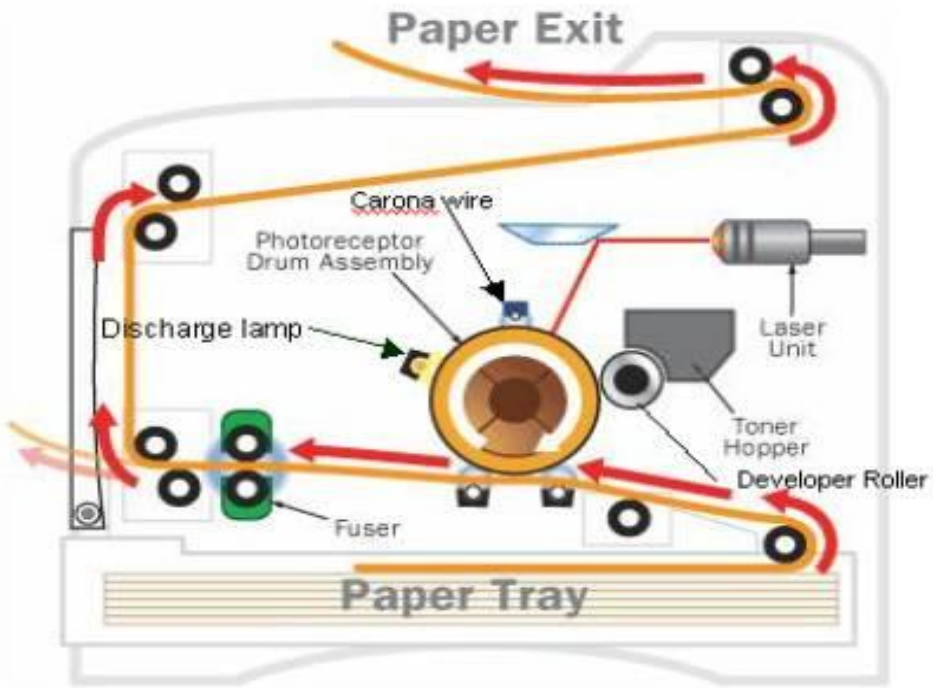
Inkjet Printers

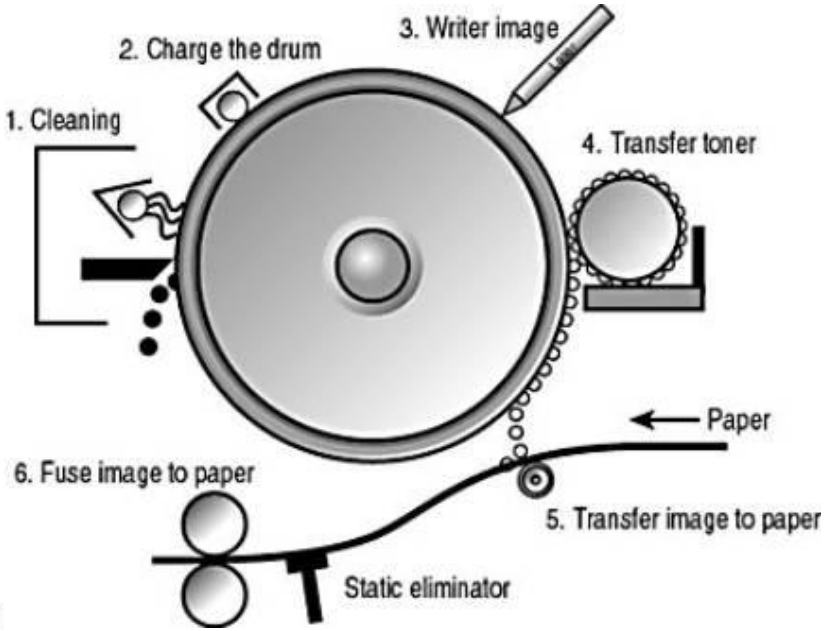
a) Edge guide - Helps load the paper straight. Adjust the left edge guide so that it fits snugly to the width of your paper.
b) Sheet feeder - Feeds a stack of paper automatically.
c) Paper support - Supports the paper loaded in the sheet feeder.
d) Feeder guard - Prevents objects placed on the document cover from falling inside the printer when opening the document cover.
e) Document cover - Open and close when you place a photo or document.
f) Output tray - Receives ejected paper.
g) Output tray extension - Supports the ejected paper.
h) Scanner unit - Open and close when you replace an ink cartridge.
i) On button - Turns the printer on and off.

Inkjet Printers

- **Advantages and Disadvantages**
  - The low cost and relatively high quality of prints that are offered by the inkjet printer is suitable for most day-to-day tasks and thus used in home and offices.
  - Inkjets have the advantage of practically no warm up time and lower cost per page, no noise
  - Inkjet printers can print on a variety of papers as they essentially work by shooting printer ink on to the paper.
  - the disadvantages are ink cartridges are expensive.

Laser Printer





Laser printer components and their functionality

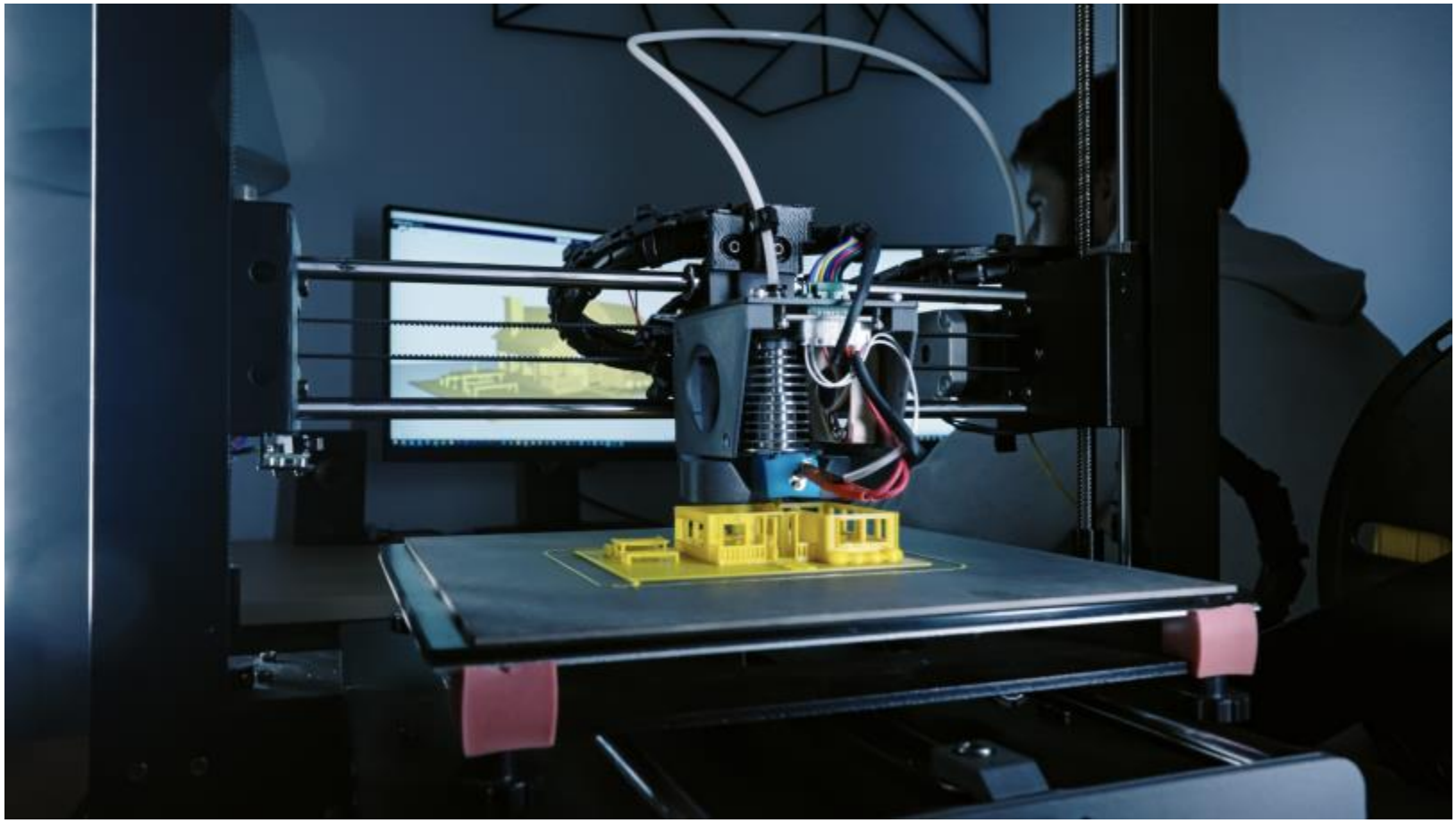
Laser Printer

1. **Paper feeding**
   - The printer moves a sheet of paper from the proper tray onto a series of rollers, through the imaging and fixing areas, and to the output hopper.
2. **Drum Cleaning and Charging**
   - Any residual toner from past jobs is scraped from the printer's photosensitive drum. A fine wire (the primary corona) produces a negative electrical charge across the entire face of the drum. The image is set in raster lines as a series of fine dots on the drum.
3. **Imaging the Drum**
   - The information from the raster-image processor is read from memory and sent to the print engine, one line at a time. The laser sets a positive charge in the areas of the image to be filled with toner.
4. **Transferring Toner to the Drum**
   - A film of fine plastic power is placed on the toner transfer roller, which is turning close to the photosensitive drum. This toner is then attracted to the positively charged areas of the drum.
5. **Transferring Toner to the Paper**
   - The corona wire places a positive electrical charge on the paper as it moves close to the drum. The toner is attracted to the page, forming an image.
6. **Fusing the Toner**
   - The page passes through a pair of rollers. The roller on the side toward the toner that has been placed on the page is heated just enough to melt the plastic toner particles onto the page without smearing. The roller on the other side supplies the needed pressure.

Laser Printer

- Laser printers have a number of advantages over the rival inkjet technology. They produce much better quality black text documents than inkjets, and they turn out more pages per minute (100 to 200 pages per minute are typical) at a lower cost per page than inkjets.
- Laser printers are well known for their speed and they can handle large volumes and another advantage is that they are not messy as inkjet that is, there is no ink spillage as the ink is created from powder and they can print on any type of paper
- the disadvantage is that laser printers are expensive.

3D Printer



3D Printer

3D PRINTING USES

- Construction
- Automotive
- PPE
- Medical Equipment
- Food
- Aerospace
- Education

The 3D Printing Process

- 3D printers are related to additive manufacturing. 3D printers use computer-aided design to understand a design. When a design is ready, a material that can be dispensed through a hot nozzle or precision tool is printed layer by layer to create a three-dimensional object from scratch.
- When the modeling and slicing of a 3D object is completed, it’s time for the 3D printer to finally take over. The printer acts generally the same as a traditional inkjet printer in the direct 3D printing process, where a nozzle moves back and forth while dispensing a wax or plastic-like polymer layer-by-layer, waiting for that layer to dry, then adding the next level. It essentially adds hundreds or thousands of 2D prints on top of one another to make a three-dimensional object.

3D Printer

- **HOW MUCH DO 3D PRINTERS COST?**
  - The cost of 3D printers vary based on the size, specialty and use. The cheapest 3D printers, for entry level hobbyists, typically range from $100 to $500. More advanced models can range between $300 and $5,000. Industrial 3D printers can cost up to $100,000.

3D Printer

- https://youtu.be/bj8kZ3llS5E

3D Printing

- **Advantages**
  - 3D Printers are Affordable
  - 3D Printers are Fast
  - 3D Printers Can Work with Speciality Materials
- **Disadvantages**
  - 3D Printers May Not Provide Enough Strength
  - 3D Printers May Have Accuracy Issues
  - 3D Printers May Require Post-Processing

Introduction to Scanners

- Scanner is a device that scans images, printed text, and handwriting etc and converts it to digital form or image.
- It is so named because the data is converted one line at a time or scanned down the page as the scanning head moves down the page.

Components inside a scanner

- Glass Plate and Cover
  - The glass plate is the transparent plate wherein the original is placed so that the scanner can scan it and the cover keeps out stray light that can affect the accuracy of the scan
- Stepper Motor
  - The stepper motor in a scanner moves the scan head down the page during scan cycle and this is often located either on the scan head itself or attached to a belt to drive the scanner head.

Components inside a scanner

- **Scanning head**
  - Scanning head is the most important component because it is the one which does actual scanning. It contains components like
  - **Light source and mirror** : It is the bright white light that is used to illuminate the original as it is being scanned and which bounces off the original and reflected off several mirrors

Components inside a scanner

- **Stabilizer bar**: It is a long stainless steel rod that is securely fastened to the case of the scanner and it provides a smooth ride as the scanner scans down the page
- **CCD (Charge Coupled Device) or CIS (Contact Image Sensor)**: A CCD array is a device that converts photons into electricity. Any scanner that uses CCD use lens to focus the light coming from the mirrors within the scanning head.
- Another technology used in some cheaper scanners is CIS wherein the light source is a set of LEDs that runs the length of the glass plate.

Types

- Flatbed Scanners
- Sheetfed Scanners
- Handheld Scanners

Flatbed Scanners

- The most commonly used scanner is a flatbed scanner also known as desktop scanner.
- It has a glass plate on which the picture or the document is placed.
- The scanner head placed beneath the glass plate moves across the picture and the result is a good quality scanned image.
- For scanning large maps or toposheets wide format flatbed scanners can be used.

Sheetfed Scanners

- Sheet fed scanners work on a principle similar to that of a fax machine.
- In this, the document to be scanned is moved past the scanning head and the digital form of the image is obtained.
- The disadvantage of this type of scanner is that it can only scan loose sheets and the scanned image can easily become distorted if the document is not handled properly while scanning.

Handheld Scanners

- Hand-held scanners although portable, can only scan images up to about four inches wide.
- They require a very steady hand for moving the scan head over the document.
- They are useful for scanning small logos or signatures and are virtually of no use for scanning maps and photographs.

Printer and Scanner Interfaces

- A printer's interface is a combination of hardware and software that allows the printer to communicate with a computer.
- The hardware interface is called port and each printer has atleast one interface. An interface incorporates several components including its communication type as well as the interface software.
- There are eight major communication types

Printer and Scanner Interfaces

1. Serial : Computers send one bit at a time, one after another through serial connection. The communication parameters like parity, baud should be set on both entities before communication takes place.

2. Parallel : Parallel communication is more popular with printers because it is more faster than serial communication. In this type of communication, printer receives eight bits at a time over eight separate wires.

Parallel uses a DB25 connection on the computer side and a oddly shaped 36 pin connection on the printer.

3. USB (Universal Serial Bus) : These can transfer data quickly with a transfer rate up to 12 Mbps and it automatically recognizes new devices.

4. Network : Network, or commonly Ethernet, connections are commonplace on network laser printers, though some other types of printers do employ this type of connection. These printers have a Network Interface Card (NIC) and ROM-based software that allows them to communicate with networks, servers and workstations.

Printer and Scanner Interfaces

5. Infrared : Infrared transmissions are wireless transmissions that use radiation in the infrared range of the electromagnetic spectrum. An Infrared acceptor allows your devices (laptops, PDA's, Cameras, etc) connect to the printer and send print commands via infrared signals.

6. SCSI (Small Computer System Interface) : Laser printer, dye-sublimation printers and few other use SCSI interfaces to PC as there are benefits like daisy chaining wherein more than one device could be on a single SCSI connection and it is easy to implement.

7. IEEE 1394 Firewire : Firewire is a high speed connection typically used for digital video editing or other high bandwidth requirements. This interface currently supports devices with a maximum throughput of 800 Mbps and capable of speeds up to 3.2 Gbps.

8. Wireless : Wireless is the currently popular technology like infrared, Bluetooth, 802.11 etc. The information is transmitted wirelessly through the air using radio waves and is received by the device.

Bluetooth is used to replace the cables between computers and its peripherals and they usually work over small distances of about 10 meters.

Out of these above communication types scanners mostly use USB, Parallel, SCSI, IEEE 1394/FireWire.

Installing And Configuring Printers And Scanners

- Step 1 : Attach the device using a local or network port and connect the power
- Step 2 : Install and Update the Device driver and calibrate the device
- Step 3 : Configure options and default settings
- Step 4 : Print/scan a test page

Printers And Scanners Problems And Troubleshooting

- Dot-Matrix printer problems
  - Print results are too light
  - Printer does not print
  - Characters are incomplete
  - Printout jams inside the printer
  - Stepper motor problems

Printers And Scanners Problems And Troubleshooting

- Laser printer problems
  - Print results too light, white/light columns
  - Ghosting
  - Prints black pages
  - Memory Overflow error
  - Paper jams
  - Repetitive small marks or defects
  - Image Smudging

Scanner Problems and Troubleshooting

- Common problems with scanner are scanners won't turn on, strange noises from scanner, Scanners won't scan etc.
- Tips to troubleshoot the basic scanner related problems are:
  - Ensure that the scanner software is properly installed and all the settings are properly configured
  - Make sure that the power light on the scanner is turned on.
  - If any software related problems are noticed in scanner then uninstall and reinstall the scanner software
  - Try to perform the basic tests that are provided by the manufacturer.
  - Make sure that the cables to the scanner are properly connected.
  - Overheating, dust and humid environment can cause permanent damage to the scanner.
  - Restarting your computer, unplugging cables and verifying any hardware conflicts in the device manager can diagnose and troubleshoot the possible errors.

Tools To Troubleshooting Problems

- Multimeter
- Screw driver
- Cleaning solutions
- Extension Magnets
- Test patterns

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Thank you